Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60 (N&S), Elite, IAS & LIIT | JEE-MAIN- Shift-1 | Date: 31-03-2026
Time: 09.00Am to 12.00Pm | GTM-34 | Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**3**	2	**4**	3	**3**	4	**2**	5	**1**
6	**3**	7	**1**	8	**3**	9	**2**	10	**2**
11	**1**	12	**2**	13	**1**	14	**1**	15	**4**
16	**3**	17	**3**	18	**2**	19	**4**	20	**3**
21	**0**	22	**15**	23	**13**	24	**0**	25	**1**

PHYSICS

26	**4**	27	**2**	28	**3**	29	**1**	30	**4**
31	**1**	32	**4**	33	**4**	34	**3**	35	**3**
36	**4**	37	**1**	38	**1**	39	**1**	40	**3**
41	**1**	42	**3**	43	**2**	44	**4**	45	**1**
46	**44**	47	**500**	48	**13**	49	**15**	50	**7**

CHEMISTRY

51	**3**	52	**4**	53	**2**	54	**2**	55	**2**
56	**3**	57	**1**	58	**1**	59	**1**	60	**2**
61	**4**	62	**2**	63	**1**	64	**2**	65	**2**
66	**1**	67	**2**	68	**1**	69	**1**	70	**1**
71	**4**	72	**4**	73	**4**	74	**36**	75	**7**

For More Join: @IITJEE_Advanced

SOLUTIONS

MATHEMATICS

1. AAAPPNNHYSKTIR

i) All 4 are different $= 10_{C_4}.4! = 5040$

ii) 3 same 1 different $9C_1 \frac{4!}{3!} = 36$

iii) 2 same 2 different $3_{C_1} 9_{C_2} . \frac{4!}{2!} = 1296$

iv) 2 same s same $3_{C_2} . \frac{4!}{2!2!} = 18$

$\therefore 5040 + 36 + 1296 + 18 = 6390 \quad = \quad \alpha + \beta = 9 \,|\, A| = 3 \quad = \quad |\, A^{-1}.(AB)^T \,| = 9$

2.

$$\int \left(\frac{1}{x^2} + \frac{1}{x^4}\right)\left(\frac{3}{x} + \frac{1}{x^3}\right)^{\frac{1}{23}} dx$$

using $t = \frac{3}{x} + \frac{1}{x^3} \Rightarrow dt = -3\left(\frac{1}{x^2} + \frac{1}{x^4}\right) dx$

$$\int \frac{t^{1/23} dt}{-3} - \frac{t^{24/23}}{\left(\frac{24}{23}\right)(-3)} + C$$

$\Rightarrow \alpha = 23 \quad \beta = -1 \quad \gamma = -3$

$\alpha + \beta + \gamma = 19$

Therefore the number of natural number of less than 19 and co-prime to it = 18

3. Put $\frac{z_1}{z_2} = z \quad \therefore \left|\frac{z_1}{z_2}\right| = |z| \Rightarrow |z| = \frac{1}{10}$

Also, $\theta = \arg\left(\frac{z_1}{z_2} - 1\right) = \arg(z-1) \quad \therefore \tan^2\theta]_{\max} = \frac{1/100}{99/100} = \frac{1}{99}.$

4.

Hyp : $\frac{y^2}{64} - \frac{x^2}{49} = 1$

An ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through the vertices of the hyperbola $H : \frac{x^2}{49} - \frac{y^2}{64} = -1.$

So $b^2 = 64$

$e_H = \sqrt{1 + \frac{a^2}{b^2}} = \sqrt{1 + \frac{49}{64}}$

Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

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$$e_E=\sqrt{1-\frac{a^2}{b^2}}=\sqrt{1-\frac{a^2}{64}}$$

$$b=8,\sqrt{\frac{1-a^2}{64}}\times\frac{\sqrt{113}}{8}=\frac{1}{2}\Rightarrow\sqrt{64-a^2}\times\sqrt{113}=32$$

$$\left(64-a^2\right)=\frac{32^2}{113}$$

$$\Rightarrow a^2=64-\frac{32^2}{113}$$

$$l=\frac{2a^2}{b}=\frac{2}{8}\left(64-\frac{32^2}{113}\right)=\frac{1552}{113}$$

$$113l=1552$$

5. Let $\overline{\beta_1}=\lambda\overline{\alpha},\ \overline{\beta_2}=\overline{\beta}-\overline{\beta_1}\quad=\left(\bar{i}+2\bar{j}-4\bar{k}\right)-\lambda\left(4\bar{i}+3\bar{j}+5\bar{k}\right)$ and $\overline{\beta_2}.\overline{\alpha}=0\rightarrow\lambda=-1/5$

$$\overline{\beta_2}=\frac{9}{5}\bar{i}+\frac{13}{5}\bar{j}-3\bar{k}\quad\overline{\beta_2}=-\frac{4}{5}\bar{i}-\frac{3}{5}\bar{j}-\bar{k}$$

$$\overline{\beta_2}-\overline{\beta_1}=\frac{13}{5}\bar{i}+\frac{16}{5}\bar{j}-2\bar{k}\rightarrow5\left(\overline{\beta_2}-\overline{\beta_1}\right)=13\bar{i}+16\bar{j}-10\bar{k}.$$

6. $$I=\int_0^1(x-f(x))^{2026}dx \qquad \ldots\ldots\ldots\ldots\ (1)$$

$$x=f(t)\Rightarrow dx=f'(t)dt$$

$$I=\int_1^0(f(t)-f(f(t)))^{2026}f'(t)dt\qquad\{\because f(f(0))=0\Rightarrow f(1)=0\}$$

$$I=-\int_0^1(f(t)-t)^{2026}f'(t)dt$$

$$I=-\int_0^1(t-f(t))^{2026}f'(t)dt\ \ldots\ldots\ldots\ (2)$$

Eqn (1) +Eqn (2)

$$2I=\int_0^1(x-f(x))^{2026}(1-f'(x))dx=\left(\frac{(x-f(x))^{2027}}{2027}\right)_0^1=\frac{1}{2027}-\left(\frac{-1}{2027}\right)$$

$$I=\frac{1}{2027}=\frac{p}{q}\Rightarrow(p+q)=2028$$

7. Let root of $x^2-6x+a=0$ are $\alpha,4\beta\Rightarrow4\alpha\beta=a,\alpha+4\beta=6$

$x^2-cx+6=0$ are $\alpha,3\beta\Rightarrow3\alpha\beta=6,\alpha+3\beta=C$

Integral roots are 2,4 and 2,3. common root is 2.

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8. Given, $T_4 = 500$ let first term =a

r=common ratio $= \frac{1}{m}, m \in N$, then, $ar^3 = 500 \Rightarrow \frac{a}{m^3} = 500$

$S_n - S_{n-1} = ar^{n-1} \Rightarrow S_6 > S_5 + 1 \Rightarrow S_6 - S_5 > 1$

$ar^5 > 1 \Rightarrow \frac{500}{m^2} > 1$

$m^2 < 500$ ……. (i)

And $S_7 - S_6 < \frac{1}{2} \Rightarrow \frac{a}{m^6} < \frac{1}{2}$

$m^3 > 10^3 \Rightarrow m > 10$ …(ii)

From (i) and (ii)

m=11,12,13…..22

i.e $\lambda = 12$, variance of 1,2,…..11 is 10

9. We have number of ways of drawing five tickets= $4 \times 4 \times 4 \times 4 \times 4 = 4^5$

So, total no of elementary events = 4^5

Now, favourable no of elementary events = no of ways of getting 24 as the sum of numbers on the five tickets.

=coefficient of x^{24} in $(x^0 + x^1 + x^{10} + x^{11})^5$ = coefficient of x^{24} in $(1+x)^5(1+x^{10})^5$

= coefficient of x^{24} in $(1+x)^5.\left(.5C_0 + .5C_1 x^{10} + .5C_2 x^{20} x^{20} + ...\right)$

$= .5C_2 \times$ coefficient of x^4 in $(1+x)^5 = .5C_2 \times .5C_4 = 50$

Hence, required probability= $\frac{50}{4^5} = \frac{25}{512}. = \frac{2^4 + 9}{512} \therefore \lambda = 4$

3^{4k} end with 1, 3^{4k+1} end with 3, 3^{4k+2} end with 9, 3^{4k+3} end with 7

$(1! + 2! + 3! + ... + \lambda!)^{2026!} = (33)^{2026!}$ end with 1

10. Term independent of x in the expansion of $\left(3x - \frac{1}{x}\right)^{20}$

$T_{r+1} = 20_{C_r} (3x)^{20-t} \left(\frac{-1}{x}\right)^r$

Where r=10

$\alpha = T_{11} = 20_{C_{10}} 3^{10}$ i.e $\alpha = 11$ …… (1)

Term independent of x in the expansion of $\left(x + \frac{\sqrt[9]{3^{10}}}{x}\right)^{18}$

$T_{r+1} = 18_{C_r} (x)^{18-t} \left(\frac{(3^{10})^{t-9}}{x}\right)^t$

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When r=9

$\beta = T_{10} = {}^{18}C_9 3^{10}$ i.e $\beta = 10$

$(\alpha+\beta)^{2026} = (11+10)^{2026} = (21)^{2026} = 3^{2026}.7^{2026}$

11. Equation of parabola, $x^2 = 12y$

$SA \perp AB$

So, $m_{As}.m_{AB} = -1$

$$\frac{\left(3-\frac{t^2}{3}\right)}{(0-2t)(}.\frac{\left(\alpha-\frac{t^3}{3}\right)}{(0-2t)} = -1 \Rightarrow 3\alpha = \frac{27t^2+t^4}{t^2-9}$$

Ordinate of centroid of $\Delta SAB = k = \frac{\alpha+\frac{t^2}{3}}{3}$

$$k = \frac{9+3\alpha+t^2}{9}$$

Take limit both sides

$$\lim_{t\to 1} k = \lim_{t\to 1}\frac{1}{9}\left(9+t^2+\frac{27t^2+t^4}{(t^2-9)}\right) = \frac{13}{18}$$

$[\lambda] = 0, \left[\frac{\lambda}{2}\right] = 0.....$

12. $\lim_{t\to x}\frac{t^{10}f(x)-x^{10}f(t)}{t^9-x^9} = 1$

$$\lim_{t\to x}\frac{10t^9 f(x)-f'(t)x^{10}}{9t^8} = 1 \Rightarrow 10x^9 f(x)-f(x)x^{10} = 9x^8 \Rightarrow f'(x)-\frac{10}{x}f(x) = -\frac{9}{x^2}$$

$IF = e^{-\int\frac{10}{x}dx} = \frac{1}{x^{10}}$

$\therefore Sol^n$

$$\frac{y}{x^{10}} = \int -\frac{9}{x^{10}}\times\frac{1}{x^2}dx = -9\int x^{-12}dx$$

$$\frac{y}{x^{10}} = \frac{y}{11}x^{-11}+C$$

$$\because y(1) = 2 \Rightarrow C = \frac{13}{11} \Rightarrow y = \frac{9}{11x}+\frac{13}{11}x^{10}$$

13. $\vec{p} = 2\hat{i}+3\hat{j}+4\hat{k}, \vec{q} = 3\hat{i}+4\hat{j}+5\hat{k} \Rightarrow \vec{p}\times\vec{q} = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & 4\\ 3 & 4 & 5\end{vmatrix} = -\hat{i}+2\hat{j}-\hat{k}$

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$A \equiv (1,2,3), B \equiv (\lambda,4,5)$

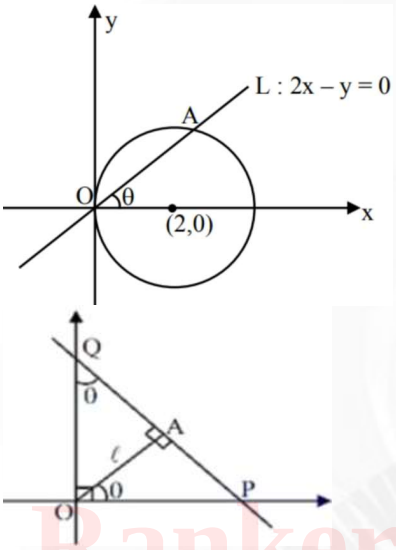
Shortest distance $= \left|\dfrac{\overrightarrow{AB}.(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}\right| \Rightarrow |\lambda-1+4-2|=1 \Rightarrow |\lambda-3|=1 \Rightarrow \lambda = 4,2$

$r = \dfrac{abc}{4\Delta} = \dfrac{5\sqrt{2}}{3}$

14. Given circle is $C_1 : x^2 + y^2 - 4x = 0, \tan\theta = 2$

C_2 is a circle with OA as diameter,

So, tangent at A and C_2 is perpendicular to OA



Therefore, $\dfrac{QA}{AP} = \dfrac{l\cot\theta}{l\tan\theta} = \dfrac{1}{\tan^2\theta} = \dfrac{1}{4}$

15. $\because f(x+y) = f(x)f(y) \Rightarrow f(x) = a^x$

$\because f(1) = 7 \Rightarrow a = 7 \Rightarrow f(x) = 7^x, \forall x \in N$

$\because g(x+y) = g(xy),\ g(1) = 1$

Put $x = y = 1 \Rightarrow g(2) = g(1) = 1$

Put $x = 1,\ y = 2 \Rightarrow g(3) = g(2) = 1$

Put $x = 1,\ y = 3 \Rightarrow g(4) = g(3) = 1$

$\Rightarrow g(x) = 1,\ \forall x \in N$

$\Rightarrow \sum_{x=1}^{n}\left(\dfrac{f(x)}{g(x)}\right) = \sum_{x=1}^{n} 7^x = \dfrac{7(7^n-1)}{7-1}$

$\Rightarrow \dfrac{7}{6}(7^n-1) = 19607$

$\Rightarrow 7^n = 16807 = 7^5 \Rightarrow n = 5$

$\Rightarrow$ Option (4) is Correct.

16. $f(x) = x^x\ (x>0) = f'(x) = x^x(1+\log_e x) = f'(x) > 0 = x^x(1+\log_e x) > 0$

$1+\log_e x > 0 = x \in \left(\dfrac{1}{e}, \infty\right)$

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17. $\sum_{n=1}^{6} \csc\left(\theta+(n+1)\frac{\pi}{4}\right).\csc\left(\theta+n\frac{\pi}{4}\right)=4\sqrt{2}$

$\sqrt{2}\left[\cot\theta+\cot\left(\theta+\frac{3\pi}{2}\right)\right]=4\sqrt{2}=\cot\theta+\tan\theta=4$

$1+\tan^2\theta=4\tan\theta=\begin{matrix}\theta=15^0,75^0\\ |\alpha-\beta|=60^0\end{matrix}$

18. $2\log_3^x+\left(\log_3^x\right)^2-10=-2$

Let $\log_3^x=k$

$k^2+2k-8=0$

$k=2,k=-4$

$x=9$ or $\frac{1}{81}$

19. Conceptual

20. Suppose E_1 : Bag A is selected

A	B
3W	3W
7R	2R

E_2 : Bag B is selected

E: white ball is drawn from the bag

$$\therefore P(E_1/E)=\frac{P(E).P(E/E_1)}{P(E_1).P(E/E_1)+P(E_2).P(E/E_2)} \qquad =\frac{\frac{1}{2}\times\frac{3}{10}}{\frac{1}{2}\times\frac{3}{10}+\frac{1}{2}\times\frac{3}{5}}=\frac{3}{3+6}=\frac{1}{3}.$$

21. $\frac{PA}{QA}=\frac{2}{3};\frac{PB}{QB}=\frac{2}{3};\frac{PC}{QC}=\frac{2}{3}$

$\therefore A,B,C$ lies on circle with diameter ends $\left(\frac{-12}{5},\frac{12}{5}\right);(-12,-12)$, his circle radius is

$\frac{12}{5}\sqrt{13}$

Required circle radius is $\frac{4}{5}\sqrt{13}$

22. equation of line passing through P(1,4,3) is $\frac{x-1}{A}=\frac{y-4}{B}=\frac{z-3}{C}$ ---(1)

Equation (1) is perpendicular to $\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{4}$ and $\frac{x+2}{3}=\frac{y-4}{2}=\frac{z+1}{-2}$

Hence 2A+B+4C=0 and 3A+2B-2C=0

$\frac{A}{-2-8}=\frac{B}{12+4}=\frac{C}{4-3}\Rightarrow\frac{A}{-10}=\frac{B}{16}=\frac{C}{1}$

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Hence equation is $\frac{x-1}{-10}=\frac{y-4}{16}=\frac{z-3}{1}$ ----- (2)

Now any point Q on equation (2) $(1-10\lambda,16\lambda+4,\lambda+3)$

Distance $(PQ)^2=(10\lambda)^2+(16\lambda)^2+\lambda^2=357$

$\lambda=\pm1$

Q(-9,20,4) or Q(11,-12,2)

Hence a+b+c=15 or 1

23. Given $R=\{(a,b),(b,c),(b,d)\}$

In order to make it equivalance relation as per given set R must be

$\{(a,a),(b,b),(c,c),(d,d),(a,b),(b,a),(b,c),(c,b),$
$(b,d),(d,b),(a,c),(a,d),(c,d),(d,c),(c,a),(d,a)\}$

There already given so 13 more to be added.

24. $\tan^{-1}\left(\frac{1}{1+n(n+1)}\right)=\tan^{-1}(n+1)-\tan^{-1}(n)$

$$\Rightarrow\sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)=\left(\tan^{-1}2-\tan^{-1}1\right)+\left(\tan^{-1}3-\tan^{-1}2\right)+\ldots+\left(\tan^{-1}51-\tan^{-1}50\right)$$

$$=\tan^{-1}51-\tan^{-1}1=\tan^{-1}\left(\frac{50}{52}\right)=\cot^{-1}\left(\frac{52}{50}\right)$$

a+b=52+50=102 is a multiple of 3

25. Solving system of equations, we get point of intersections (1,2), (7,5), (2,3)

Area of $\Delta DEF=\frac{1}{2}\begin{vmatrix}1&2&1\\7&5&1\\2&3&1\end{vmatrix}=\frac{1}{2}|2-10+11|=\frac{1}{2}(3)=\frac{3}{2}$

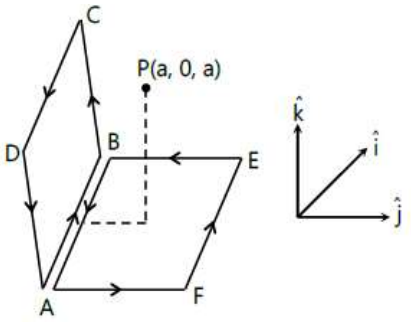
$ar\Delta ABC=4ar\Delta DEF=4\left(\frac{3}{2}\right)=6sq.unit.$

$$\int_0^{3/2}\left[x^2\right]dx=\int_0^1\left[x^2\right]dx+\int_1^{\sqrt{2}}\left[x^2\right]dx+\int_{\sqrt{2}}^{3/2}\left[x^2\right]dx=(0)_0^1+(x)_1^{\sqrt{2}}+(2x)_{\sqrt{2}}^{3/2}=2-\sqrt{2}$$

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PHYSICS

26. resultant magnetic field at P will be $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{k}\right)$



27. Diode is in reverse bias hence $i_1 = 0$ and $i_2 = 0.2\text{A}$

28. $R_{BC} = R_{AB} = R_{CB}, \left(\frac{Q}{t}\right)_{AB} = \left(\frac{Q}{t}\right)_{AD}$

29. balanced weatstone's bridge

30. $S_A = \frac{1}{2}\times 60\times 18$, $S_B = 40\times 18$, $\Delta S = S_A - S_B$

31. $\frac{\alpha z}{k\theta} = 1, \alpha = \frac{k\theta}{z}, p = \frac{\alpha}{\beta}, \beta = \frac{\alpha}{p}$

32. First law is fundamental principle that defines inertial frame.
Without first law defining frame, second law is ambiguous.

33. $W - Wg + Ws = \Delta k.E = 0$, $\Delta U = W + E$
W by spring may be positive or negative depending on displacement

34. $\Delta E = -13.6\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), E = \frac{hc}{\lambda}$

35. Conceptual

36. If current in both inductor is in same direction then $L_{eq} = L_1 + L_2 + 2M$ and, if current in both inductor is in opposite direction, then $L_{eq} = L_1 + L_2 - 2M$

37. $\rho_t = \rho_0(1+\alpha t)$

38. (A) $I_t = \frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$

(B) $I = I_{cm} + MR^2 = \frac{2}{3}MR^2 + MR^2 = \frac{5}{3}MR^2$

(C) $I = \frac{MR^2}{2}$ (D) $I = \frac{MR^2}{4}$

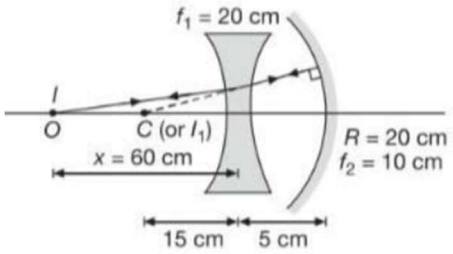
39. $\bar{E} = E_0\hat{n}\sin\left(\omega t - \bar{k}.\bar{r}\right)$ Wave vector $\bar{k} = -6\hat{j} + 8\hat{k}$ $\hat{s} = \frac{\bar{k}}{|\bar{k}|}$

40. $i_1 = \frac{V_{AB}}{G} = 4 \Rightarrow V_{AB} = 4G$ $i_2 = \frac{V_{AB}}{G+3} = 2 \Rightarrow \frac{4G}{G+3} = 2$
$G = 3\Omega$ $V_{AB} = i_1 G = 4\times 3 = 12V$

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41. $V_S \propto f$, $i \propto I$

42.



$$\Rightarrow \frac{1}{\upsilon}-\frac{1}{u}=\frac{1}{f} \qquad \Rightarrow -\frac{1}{15}+\frac{1}{x}=\frac{-1}{20}$$

$$\Rightarrow \frac{1}{x}=\frac{1}{15}-\frac{1}{20}=\frac{4-3}{60}=\frac{1}{60} \qquad \Rightarrow x=60 \text{ cm}$$

43. V on each capacitor = 2.5 V $\qquad V_{AX}=3\times 2.5=7.5V$

44. Applying energy conservation form P to O

$K_1+U_k=K_1+U_2$

$$O+(M)\left(\frac{-GM}{\sqrt{R^2+(2R)^2}}\right)=\frac{1}{2}m_0V^2+(m_0)\left(\frac{-GM}{R}\right) \quad V=\sqrt{\frac{2GM}{R}\left(1-\frac{1}{\sqrt{5}}\right)}$$

45. $q_{B_1}=-2cE \quad V_1=V_2 \Rightarrow \frac{q_{1f}}{C}=\frac{q_{2f}}{C} \qquad q_{2f}=-CE$

46. $X_C=\frac{1}{\omega C}=0,\ X_L=\omega L=\infty,\ I=\frac{220}{5}=44A$

47. $V=\sqrt{V_R^2+(V_L-V_C)^2}$

48. $Q=Q_{BC}+Q_{CD} \qquad =nC_v(T_C-T_B)+nC_p(T_C-T_D) \qquad \frac{3}{2}P_0V_0+\frac{5}{2}(2P_0V)$

49. $\frac{20\lambda D}{d}=\frac{2\lambda D}{a}\Rightarrow a=\frac{d}{10}$

50. At $x=\frac{2A}{3} \qquad v=\omega\sqrt{A^2-\left(\frac{2A}{3}\right)^2}=\frac{\sqrt{5}A\omega}{3}$

Let A' be the new amplitude of particle when speed is increased to three times.

$$\therefore 3v=\sqrt{5}A\omega=\omega\sqrt{(A')^2-\left(\frac{2A}{3}\right)^2}$$

$$\Rightarrow 5A^2=(A')^2-\frac{4A^2}{9} \qquad \Rightarrow (A')^2=\frac{49A^2}{9}\Rightarrow A'=-\frac{7A}{3}=\frac{nA}{3}(\text{given})$$

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CHEMISTRY

51. Conceptual

52. At low pressure $\frac{-dp}{dt}=\frac{+k^1kp}{1}[\because p\approx 0]$ $\frac{-dp}{dt}=k^{11}[p]^1$

First order at low pressure.

At high pressure $\frac{-dp}{dt}=\frac{k^1kp}{kp}[k1\ kp>>\theta]$

Zero order at high pressure

53. $\left[Cr(H_2O)_6\right]^{2+};\left[Fe(H_2O)_4\right]^{2-}$ $Cr^4\rightarrow d^4$ $Fe^{2+}\rightarrow d^6$

4-unpaired $e^{(-)}s$ 4-unpaired $e^{(-)}s$

54. $x=\left[Cr(NH_3)_6\right]^{3+};y=\left[Ni(CN)_4\right]^{2-}$

$Cr^{3+}\rightarrow d^3$ $Ni^{2+}\rightarrow d^8$

$Cr^{3+}\rightarrow inpaired\ \ d^2sp^3;Ni^{2+}\rightarrow dsp^2$

55. Conceptual

56. Conceptual

57. Conceptual

58. $Cl-1s^22s^2sp^63s^23p^5$

$F=1s^22s^2sp^5$

$Cl>F$

$N=1s^22s^22p^3$

$Na-1s^22s^22p^63s^1$

$Na<N<F<Li$

59. If 0.4 F electricity supplies

At cathode

$Cu^{2+}+2e^-\rightarrow Cu_{(s)}$

$0.1mole$ $0.4F$ $0.1mole$

At anode

$2H_2O_{(l)}\rightarrow O_2\uparrow+4H^++4e^{(-)}$

0.1 mole 0.4 mole 0.4 F

$V_{gas}at\ STP=(0.1+0.1)$

$=0.2\times 22.4\ =4.48Lit$

$H^++OH^-\rightarrow H_2O$

0.4 0.2 0.2 mole

-0.2 mol

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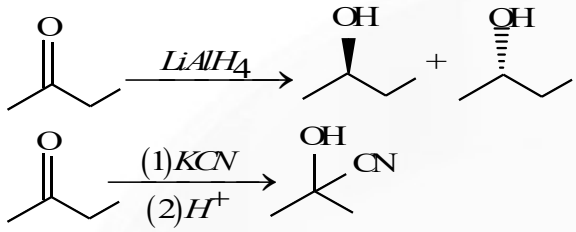
$$\left[H^+\right]_2 = \frac{0.2}{1} = 0.2M$$

$$P^H = 1 - 0.3 = 0.7$$

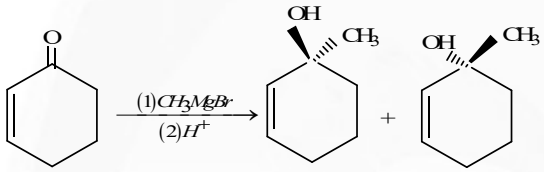
60. Conceptual

61. Chiral, optically active contains C_3-axis of symmetry

62.



$$Ph - CH_2 - Cl \xrightarrow{KCN} Ph - CH_2 - CN - Cl^-$$



63. E_{photon} = work function +kinetic energy

E_{photon} = Threshold frequency+ kinetic energy

$\therefore hv_1 = hv_0 + KE_1 \ldots\ldots\ldots(1)$

$and \;\; hv_2 = hv_0 + KE_2 \ldots\ldots\ldots(2)$

Multiply Eq(1) by 2 subtract from Eq(2)

$2hv_1 = hv_0 + KE_1 \qquad (\therefore 2KE_1 = KE_2)$

$$v_0 = 2v_1 - v_2 = \left[\frac{2c}{\lambda_1} - \frac{c}{\lambda_2}\right]$$

$$v_0 = \frac{3\times10^8}{10^{-10}}\left[\frac{2}{2200} - \frac{1}{1900}\right] \qquad \left(\therefore \left(1\overset{0}{A} = 10^{-10}m\right)\right)$$

$$v_0 = 1.1483\times10^{15}\,\text{sec}^{-1}$$

$$\text{Also, } \lambda_0 = \frac{c}{v_0} = \frac{3\times10^8}{1.148\times10^{15}} = 2.6126\times10^{-7}m$$

65. $H_2S = 2H^+ + s^{2-}$

$$K_{a_1}\times Ka_2 = \frac{\left[H^+\right]^2\left[S^{2-}\right]}{[Hs]} \qquad \Rightarrow 10^{-20}\times0.10 = \left[H^+\right]\left[S^{2-}\right]$$

$$ZnS \rightleftarrows Zn^{2+} + S^{2-}$$

$$K_{sp} = 10^{-21}$$

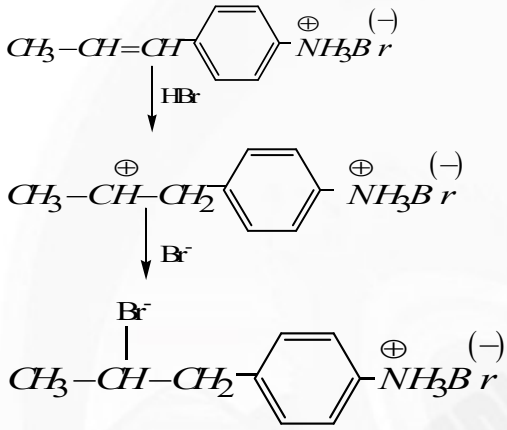
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$10^{-21} = (0.1)\left[S^{2-}\right] = \quad \left[S^{2-}\right] = 10^{-20}$

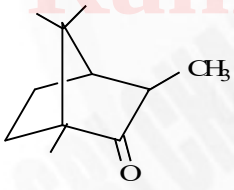
$10^{-20} \times 0.10 = \left[H^{+}\right]^{2} \times 10^{-20}$

$\left[H^{+}\right] = 0.3 = \quad P^{H} = 0.53$

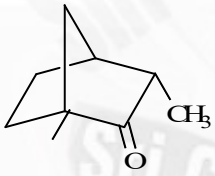
66. Conceptual
67. Conceptual
68. Conceptual
69. Conceptual
70.



71. a,b,c,e (Integer)
72. b,c,e,h (Integer)
73.



and its mirror image



and its mirror image

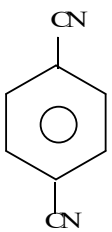
74. $4KMnO_4 + 10H_2S + 6H_2SO_4 \rightarrow 2K_2SO_4 + 4MnSO_4 + 10s + 16H_2O$

$x = 16$ (moles of H_2O produced)

$y = 28.8 = 20$ (no of $e^{(-)}s$ involved

$x + y = 16 + 20 = 36$

75. $H_2, CO_2, BF_3, CH_4, SiF_4$



BeF_2,

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